

Simple/Multi-Step Equations

Date _____ Period _____

Solve each equation.

Mild - (1-14)

Medium - (15-30)

Spicy - (31-48)

1) $3 = \frac{m}{12}$

{36}

2) $-3 = \frac{r}{12}$

{-36}

3) $-14 - x = 3$

{-17}

4) $-13 = \frac{v}{2}$

{-26}

- 5) Your grandmother gave you \$16 with which to buy a present. This covered $\frac{2}{3}$ of the cost. How much did the present cost?

\$24

- 6) If the weight of a package is multiplied by $\frac{2}{5}$ the result is 10 pounds. Find the weight of the package.

25

- 7) Mei and her best friend found some money in an envelope. They split the money evenly, each getting \$17. How much money did they find?

\$34

- 8) A stray dog ate 15 of your muffins. That was $\frac{3}{4}$ of all of them! How many are left?

5

- 9) Last Friday Eugene had \$11.90. Over the weekend he received some money for babysitting. He now has \$27.37. How much money did he receive?

\$15.47

- 10) Brenda won 65 pieces of gum playing hoops at her school's game night. Later, she gave one to each of her friends. She only has 38 remaining. How many did she give away?

27

- 11) Shanice spent \$49 on seven toy cars. How much did each toy car cost?

\$7

- 12) Your grandmother gave you \$6 with which to buy a present. This covered $\frac{2}{7}$ of the cost. How much did the present cost?

\$21

13) Ted won 45 pieces of gum playing hoops at the county fair. At school he gave one to every student in his math class. He only has 17 remaining. How many did he give away?

28

15) $\frac{a}{6} - 3 = -6$

$\{-18\}$

17) $18 = -4 - 2v$

$\{-11\}$

19) $1 = -5 + \frac{v}{3}$

$\{18\}$

21) Krystal had some candy to give to her four children. She first took three pieces for herself and then evenly divided the rest among her children. Each child received four pieces. With how many pieces did she start?

19

23) Half of your baseball card collection got wet and was ruined. You bought 13 cards to replace some that were lost. How many did you begin with if you now have 29?

32

25) Half of your baseball card collection got wet and was ruined. You bought 8 cards to replace some that were lost. How many did you begin with if you now have 20?

24

14) Last Friday Bill had \$38.63. Over the weekend he received some money for washing the dog. He now has \$57.40. How much money did he receive?

\$18.77

16) $-4(9 + p) = -84$

$\{12\}$

18) $-10 + \frac{n}{8} = -8$

$\{16\}$

20) $3a + 9 = 51$

$\{14\}$

22) Asanji won 70 lollipops playing basketball at his school's game night. Later, he gave three to each of his friends. He only has 1 remaining. How many friends does he have?

23

24) 352 students went on a field trip. Eight buses were filled and 24 students traveled in cars. How many students were in each bus?

41

26) For a field trip 21 students rode in cars and the rest filled ten buses. How many students were in each bus if 551 students were on the trip?

53

27) For a field trip 24 students rode in cars and the rest filled ten buses. How many students were in each bus if 434 students were on the trip?

41

29) Krystal had some candy to give to her five children. She first took one piece for herself and then evenly divided the rest among her children. Each child received two pieces. With how many pieces did she start?

11

31) $20 = 1 + 5a + 4$

{3}

33) $8(x + 7) = 96$

{5}

35) $-4(3x - 8) = 104$

{-6}

37) $-8n + 28 = 4(1 - 3n)$

{-6}

39) $39 + n = 4(7n + 3)$

{1}

41) $-13 = -(5 - 7b) - 8(b + 1)$

{0}

43) $5(-2n + 8) - 2(n - 2) = -28$

{6}

45) $-5(8 + 8n) - 8(-7n + 1) = 7n - 3n$

{4}

47) $2(3 - a) = 8a + 6(1 - 4a)$

{0}

28) The sum of three consecutive numbers is 57. What is the smallest of these numbers?

18

30) For a field trip 8 students rode in cars and the rest filled eight buses. How many students were in each bus if 368 students were on the trip?

45

32) $12 = -a + 5a$

{3}

34) $86 = 8(k + 7) - 2k$

{5}

36) $-5(1 - 2x) = -85$

{-8}

38) $3n - 23 = -2(1 - 5n)$

{-3}

40) $-23 + a = -6(-a + 8)$

{5}

42) $3 = -(5b - 7) + 8(b + 1)$

{-4}

44) $-6(x + 4) + 6(x - 8) = 8$

No solution.

46) $6(7 + 2k) = 3(-2 + 5k) + 3k$

{8}

48) $4k - 8(4 + k) = -8(k + 5)$

{-2}